

Sharif University of Technology

Engineering Probability and Statistics Course Project

The Coupling Method: Building a Bridge Between Random Worlds

Ali Javidan Pak, Mehdi Zekavat, Ali Ghorbani, Mohammad Ehsan Moradi

Abstract

This report presents an overview of the Coupling Method, a powerful technique in probability theory used to compare and relate different random processes. The method is particularly useful in proving convergence, ergodicity, and constructing bounds for Markov chains. This document provides a step-by-step derivation and explanation of the method's foundations and applications.

Contents

1 Problem: How different are two distributions? . . . 3

1.1 Objective: Learning how to measure the difference between two (discrete) distributions using the **Total Variation Distance**. . . . 3

1.2 Introduction . . . 3

1.3 To-Do . . . 3

2 Solution: Building a bridge using Coupling . . . 4

2.1 Objective: Understanding Coupling and its key features. . . . 4

2.2 Introduction . . . 4

2.3 To-Do . . . 5

3 Application: From Strong Secrecy to Perfect Secrecy . . . 8

3.1 Objective: Using the coupling method to prove a fundamental result in Cryptography science. . . . 8

3.2 Introduction . . . 8

3.3 To-Do . . . 8

4 Additional topic: Coupling and convergence of Markov chains . . . 10

4.1 Objective: observing an application of coupling in understanding the long-term behavior of an stochastic process. . . . 10

4.2 Introduction . . . 10

4.3 To-Do . . . 10

Appendix: Jupyter Notebook for Simulations . . . 14

1 Problem: How different are two distributions?

1.1 Objective: Learning how to measure the difference between two (discrete) distributions using the **Total Variation Distance**.

1.2 Introduction

Suppose we have two biased (loaded) dice A and B; how can we measure the "difference" in the results of these dice by reporting a single number? A natural way of achieving this, is through defining the Total Variation Distance (TVD).
For two discrete distributions p and q on the space of outcomes Ω, we define the TVD to be:

$$\|p-q\|_{\text{TV}} = \frac{1}{2}\sum_{x\in\Omega}|p(x)-q(x)| \tag{1}$$

This distance measures the maximum possible difference for the occurrence probability of an event between the two distributions and attains a value of 0 for identical distributions and 1 for non-overlapping distributions.

1.3 To-Do

1. **Theoretical Understanding**: prove the TVD formula in equation (1) is equivalent to the following formula:

$$\|p-q\|_{\text{TV}} = \sum_{x\,\text{s.t.}\,p(x)>q(x)}(p(x)-q(x)) \tag{2}$$

We'll derive the desired formula directly from equation (1) by partitioning the space Ω into two parts: part I which contains all x such that $p(x) > q(x)$ and part II which contains all x such that $p(x) \leq q(x)$. Now we may break the summation in equation (1) in the following way:

$$\begin{aligned}\|p-q\|_{\text{TV}} &= \frac{1}{2}\sum_{x\in\Omega}|p(x)-q(x)| \\ &= \frac{1}{2}\left(\sum_{\text{x}\in\text{I}}(p(x)-q(x)) + \sum_{\text{x}\in\text{II}}(q(x)-p(x))\right)\end{aligned}$$

since in part II $p(x) \leq q(x)$ and thus, $|p(x)-q(x)|$ becomes $(q(x)-p(x))$.
Now since p are q are valid distributions, we have the following normalization constraints on them:

$$\sum_{x\in\Omega}p(x)=1 \quad \text{and} \quad \sum_{x\in\Omega}q(x)=1$$

By breaking the summations we may write:

$$\begin{aligned}\sum_{\text{x}\in\text{I}}p(x)+\sum_{\text{x}\in\text{II}}p(x)=1 \quad &\Rightarrow \quad \sum_{\text{x}\in\text{II}}p(x)=1-\sum_{\text{x}\in\text{I}}p(x) \\ \sum_{\text{x}\in\text{I}}q(x)+\sum_{\text{x}\in\text{II}}q(x)=1 \quad &\Rightarrow \quad \sum_{\text{x}\in\text{II}}q(x)=1-\sum_{\text{x}\in\text{I}}q(x)\end{aligned}$$

Hence, we can substitute the expressions for $\sum_{\text{x}\in\text{II}} p(x)$ and $\sum_{\text{x}\in\text{II}} q(x)$ and get:

$$\begin{aligned}
\|p-q\|_{\text{TV}} &= \frac{1}{2}\sum_{x\in\Omega}|p(x)-q(x)| \\
&= \frac{1}{2}\left(\sum_{\text{x}\in\text{I}}(p(x)-q(x)) + \sum_{\text{x}\in\text{II}}(q(x)-p(x))\right) \\
&= \frac{1}{2}\left(\sum_{\text{x}\in\text{I}}(p(x)-q(x)) + 1 - \sum_{\text{x}\in\text{I}} q(x) - 1 + \sum_{\text{x}\in\text{I}} p(x)\right) \\
&= \frac{1}{2}\left(2\sum_{\text{x}\in\text{I}}(p(x)-q(x))\right) \\
&= \sum_{x\,\text{s.t.}\,p(x)>q(x)}(p(x)-q(x)) \qquad \text{Q.E.D.}
\end{aligned}$$

For instance, a TVD of 0.3 tells us that the distributions are not quite identical and also, not completely non-overlapping; more quantitatively, it tells us that the distributions differ at most 0.3 in probability for any event.

2. **Hands-on Calculation**: Assume the distribution p and q on the set $\Omega = \{1, 2, 3, 4\}$ are given by:

$$\begin{aligned}
p &= \{p(1) = 0.1,\, p(2) = 0.2,\, p(3) = 0.3,\, p(4) = 0.4\} \\
q &= \{q(1) = 0.4,\, q(2) = 0.3,\, q(3) = 0.2,\, q(4) = 0.1\}
\end{aligned}$$

Calculate the TVD in this case.

The calculation is straightforward. Using the formula given in equation (2) and noting that only for $x = 3$ and $x = 4$ the corresponding value $(p(x) - q(x))$ is positive, we have:

$$\|p-q\|_{\text{TV}} = \sum_{x\,\text{s.t.}\,p(x)>q(x)}(p(x)-q(x)) = (p(3)-q(3)) + (p(4)-q(4)) = 0.4$$

3. **Simulation**:

- write a function `calculate_tv_distance(p, q)` that takes two distributions as vectors (or arrays) and return their TVD. Using this function, verify the calculation in the previous section.

  The value obtained from the function is 0.4009 which is indeed very close to 0.4.

2 Solution: Building a bridge using Coupling

2.1 Objective: Understanding Coupling and its key features.

2.2 Introduction

A **Coupling** of two distributions $p(x)$ and $q(y)$ (defined on the same sapce of outcomes Ω) is a **Joint distribution** $s(x, y)$ for a pair of random variables (X, Y) such that the marginal probability mass function (PMF) of X is p and the PMF of Y is q; that is:

- $\sum_{y\in\Omega} s(x,y) = p(x) \quad ; \quad \forall x \in \Omega$
- $\sum_{x\in\Omega} s(x,y) = q(y) \quad ; \quad \forall y \in \Omega$

$s(x,y)$ can be considered as a "bridge" that connects the "separate" worlds p and q and allows us to analyze them in a single statistical framework.

2.3 To-Do

1. **The Coupling Inequality**: An essential corollary referred to as the "The Coupling Inequality" states that for any coupling (X,Y) of p and q, we have the following inequality:
$$\mathbf{P}[X \neq Y] \geq \|p-q\|_{\mathrm{TV}} \tag{3}$$
Prove the above inequality.

   Before presenting the proof, let us mention two lemmas which we will construct the proof upon; Firstly $\mathbf{P}[X \neq Y] = 1 - \mathbf{P}[X = Y]$ which is a direct consequence of the normalization condition on the distributions and secondly, $\mathbf{P}[X = z, Y = z] \leq \min(\mathbf{P}[X = z], \mathbf{P}[Y = z])$ which we will show now.

$$\{X = z, Y = z\} \subseteq \{X = z\} \Rightarrow \mathbf{P}[X = z, Y = z] \leq \mathbf{P}[X = z]$$
$$\{X = z, Y = z\} \subseteq \{Y = z\} \Rightarrow \mathbf{P}[X = z, Y = z] \leq \mathbf{P}[Y = z]$$

   From the two equations, we can deduce that:

$$\mathbf{P}[X = z, Y = z] \leq \min(\mathbf{P}[X = z], \mathbf{P}[Y = z]) \qquad \text{Q.E.D.}$$

   Now back to proving the inequality. Let us start with the expression for $\mathbf{P}[X = Y]$:

$$\begin{aligned}\mathbf{P}[X = Y] = \sum_{z\in\Omega} \mathbf{P}[X = z, Y = z] \quad &\leq \sum_{z\in\Omega} \min(\mathbf{P}[X = z], \mathbf{P}[Y = z]) \\ &= \sum_{z\in\Omega} \min(p(z), q(z))\end{aligned}$$

   By breaking the summation (as we did in section 1.3), we can write:

$$\begin{aligned}= \sum_{z\in\Omega} \min(p(z), q(z)) \quad &= \sum_{\mathrm{z\in I}} \min(p(z), q(z)) + \sum_{\mathrm{z\in II}} \min(p(z), q(z)) \\ &= \sum_{\mathrm{z\in I}} q(z) + \sum_{\mathrm{z\in II}} p(z)\end{aligned}$$

   Adding and subtracting the term $p(z)$ in the first summation and substituting for the second summation, we obtain:

$$\begin{aligned}&= \sum_{\mathrm{z\in I}} q(z) + \sum_{\mathrm{z\in II}} p(z) \\ &= \sum_{\mathrm{z\in I}} [(q(z) - p(z)) + p(z)] + 1 - \sum_{\mathrm{z\in I}} p(z) \\ &= 1 - \sum_{\mathrm{z\in I}} (p(z) - q(z)) \\ &= 1 - \|p-q\|_{\mathrm{TV}}\end{aligned}$$

Therefore, we may write:

$$\begin{aligned}\mathbf{P}\left[X \neq Y\right] &= 1 - \mathbf{P}\left[X = Y\right] \\ &\geq 1 - \sum_{z \in \Omega} \min\left(p(z), q(z)\right) \\ &= 1 - \left(1 - \|p - q\|_{\mathrm{TV}}\right)\end{aligned}$$

Hence:

$$\mathbf{P}\left[X \neq Y\right] \geq \|p - q\|_{\mathrm{TV}} \qquad \text{Q.E.D.}$$

2. **Optimal Coupling**: The algorithm below provides a method of sampling from an "optimal coupling" such that the inequality becomes an equality. In this coupling, the goal is to achieve $X = Y$ as best as we can.

Optimal sampling algorithm

(a) Compute $C = \sum_{z \in \Omega} \min\left(p(z), q(z)\right)$.

We already did this in the previous section!

$$C = \sum_{z \in \Omega} \min\left(p(z), q(z)\right) = 1 - \|p - q\|_{\mathrm{TV}} \tag{4}$$

(b) Generate a uniform random number U on the interval $[0, 1]$.

(c) If $U < C$:

- Set $X = Y$.
- Choose a value of z from the new distribution $p_{\mathrm{matched}}(z) = \frac{\min(p(z), q(z))}{C}$.
- Set $X = z$ and $Y = z$.

(d) If $U \geq C$:

- Set $X \neq Y$.
- Choose a value of x from the "remainder" $p_{rem}(x) = \frac{p(x) - \min(p(x), q(x))}{1 - C}$.
- Choose a value of y from the "remainder" $q_{rem}(y) = \frac{q(y) - \min(p(y), q(y))}{1 - C}$.
- Set $X = x$ and $Y = y$.

Explain why the probability of the event $X = Y$ in this algorithm is exactly equal to $1 - \|p - q\|_{\mathrm{TV}}$.

We may show this by employing the law of total probability and conditioning on the possible outcomes of the comparison $U \bigcirc C$:

$$\begin{aligned}\mathbf{P}[X = Y] &= \underbrace{\mathbf{P}[X = Y | U < C]}_{1} \cdot \mathbf{P}[U < C] + \underbrace{\mathbf{P}[X = Y | U \geq C]}_{0} \cdot \mathbf{P}[U \geq C] \\ &= \mathbf{P}[U < C]\end{aligned}$$

Using the probability distribution function (PDF) of a uniform random variable on the interval $[0, 1]$, we can calculate $\mathbf{P}[U < C]$:

$$\mathbf{P}[U < C] = \int_0^C \mathrm{d}u = C = 1 - \|p - q\|_{\mathrm{TV}}$$

Hence:

$$\mathbf{P}[X = Y] = 1 - \|p - q\|_{\mathrm{TV}} \qquad \text{Q.E.D.}$$

3. **Simulating the Optimal Sampling**:

- Construct a function `sample_from_optimal_coupling(p, q, n_samples)` which implements the above algorithm to generate `n_samples` pairs (x, y).
- Using the distributions for p and q given in section (1.3.2), generate 10000 samples.
- **Verification 1 (marginal distributions)**: Plot the histogram of the generated samples of x and y. Compare it to the bar chart of the original distributions p and q. Do they match?

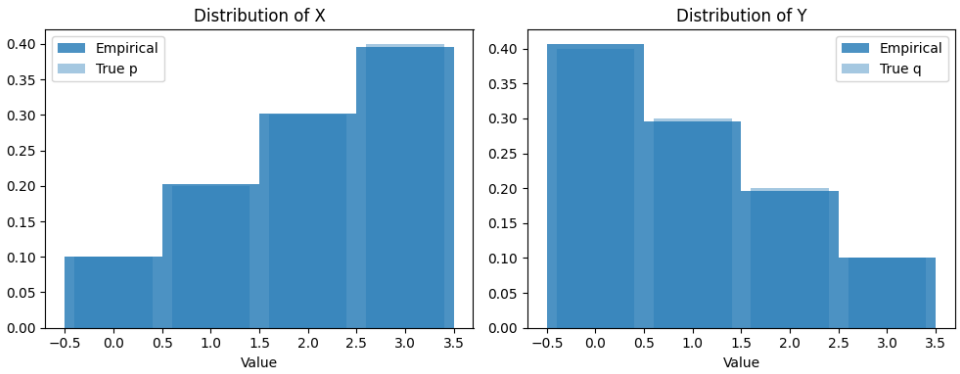


Figure 1: The histogram of the generated samples

The matching is not perfect but it is evidently a close one.

- **Verification 2 (Moment of discovery!)**: Find the empirical probability $\mathbf{P}[X \neq Y]$ from the samples $(\frac{\text{count}(x \neq y)}{\text{n}_{\text{samples}}})$ and compare it to the TVD you calculated in section (1.3.2). They should be pretty close! Explain why this result validates the optimal coupling algorithm.

  The objective of the algorithm is to achieve $X = Y$ as best as we can. Theoretically, the probability of the event $X = Y$ was found to be greater than or equal to $1 - \|p - q\|_{\text{TV}}$. Since the objective is to achieve a large value for the probability of the event $X = Y$, we should search for a way to minimize the probability of its "complement"; that is, the probability of the event $X \neq Y$ But the smallest possible value for the probability of the event $X = Y$ is determined by the Coupling Inequality and hence, we should try to bring $\mathbf{P}[X \neq Y]$ as close as possible to the value $\|p - q\|_{\text{TV}}$, which is exactly what this algorithm does! Evidently, the results from the simulations confirm the algorithm presented for optimal sampling.

3 Application: From Strong Secrecy to Perfect Secrecy

3.1 Objective: Using the coupling method to prove a fundamental result in Cryptography science.

3.2 Introduction

In a cryptography system, message M is transformed into a ciphertext C. An adversary sees C but does not have access to M.

- **Perfect Secrecy:** A system ha perfect secrecy if the ciphertext C reveals no information about the message M. In mathematical terms, M and C are independent: $P(M, C) = P(M)P(C)$.
- **Strong Secrecy:** Most practical systems attain strong secrecy rather than perfect secrecy. In this case, M and C are "almost" independent; this "proximity" is measured with the TVD: $\|P_{MC} - P_M P_C\| \leq \delta$ where δ, is a small number.
  The question is: Can a system with strong secrecy be turned into one with perfect secrecy? The answer turns out to be yes, using the coupling method!

3.3 To-Do

Suppose we have a system (Encoder 1, Decoder 1) with the following properties:

1. **Strong secrecy:** $\|P_{MC} - P_M P_C\| \leq \delta$.
2. **Reliability:** the probability of error in encoding is low: $P(\text{Decoder1}(C) \neq M) \leq \varepsilon$.

We design a new system (Encoder 2, Decoder 2). To encode the message m, the Encoder 2 operates as follows:

- We use the optimal coupling sampler from section (2.3.2).
- We consider the distributions $p = P_{MC}$ and $q = P_M P_C$.
- We then generate a pair $((m, c), (m', c'))$ from the optimal coupling of p and q.
- **Key Point: We send c' as the new ciphertext.**
- The new decoder is the same as the original decoder: $\text{Decoder}\,2\,(c') = \text{Decoder}\,1\,(c')$.

By answering the questions below, prove the efficiency of the new system:

1. **Perfect secrecy:** why does the new system have perfect secrecy?

   We consider the optimal coupling between the distributions below:

   $$P = P_{MC} \qquad\qquad Q = P_M P_C$$

   The optimal coupling produces a random pair $((M, C), (M', C'))$ such that $(M, C) \sim P_{MC}$ and $(M', C') \sim P_M P_C$. Since M' and C' are obtained from M (the original

message distribution) and C (the original ciphertext distribution) respectively, they are independent random variables; written out explicitly, we have:

$$P_{M'C'}(m', c') = Q(m', c') = P_M(m')P_C(c')$$

This confirms that M' and C' are independent (because the left-hand side is a product of two independent parts that depend only on m' and c' respectively). Since the joint distribution is the product of the marginal distributions, we have:

$$P\left(M' = m', C' = c'\right) = P\left(M' = m'\right) P\left(C' = c'\right)$$

We deduce that M' and C' are independent: $M' \perp C'$. This independence is exactly the condition required for a perfect secrecy and therefore, We have shown that the new system has perfect secrecy.

2. **Connection to Coupling:** what is the probability that the original pair (m, c) and the new pair (m', c') be different? i.e., what is $P\left[(M, C) \neq (M', C')\right]$?

   We may directly apply the result we derived in section (2.3.2) from the optimal coupling algorithm; By taking (M, C) as X and (M', C') as Y, we will have:

$$P\left[(M, C) = (M', C')\right] = C = 1 - \|p - q\|_{\mathrm{TV}} \Rightarrow P\left[(M, C) \neq (M', C')\right] = \|p - q\|_{\mathrm{TV}}$$

3. **Cost analysis:** This transformation is not cost-free and it is possible that the probability of error in decryption increases. The event corresponding to this new error is Decoder $(2)\, C' \neq M$. Show that new error is bounded above by $\delta + \varepsilon$; that is, show that $P(\mathrm{error}_{\mathrm{new}}) \leq \delta + \varepsilon$.

   The error in the new system occurs when the new decoder fails to recover the original message correctly:

$$\mathsf{error}_{\mathsf{new}} = \{\mathsf{Decoder2}(C') \neq M\}.$$

   Now we employ the law of total probability to calculate the probability of error in the new system. We split the sample space into two disjoint events:

   - Case A (Matched pairs): $(M, C) = (M', C')$ — the original and new pairs are identical.
   - Case B (Unmatched pairs): $(M, C) \neq (M', C')$ — the original and new pairs differ.

   Applying the law of total probability:

$$P(\mathsf{error}_{\mathsf{new}}) = P(\mathsf{error}_{\mathsf{new}} \mid A) \cdot P(A) + P(\mathsf{error}_{\mathsf{new}} \mid B) \cdot P(B)$$

   If $(M, C) = (M', C')$, then

$$\mathsf{Decoder2}(C') = \mathsf{Decoder1}(C') = \mathsf{Decoder1}(C),$$

   and since $M' = M$, the error probability is the same as the original system:

$$P(\mathsf{error}_{\mathsf{new}} \mid A) = P(\mathsf{Decoder1}(C) \neq M) \leq \varepsilon,$$

where ε is the maximum decoding error probability of the original system.
If $(M, C) \neq (M', C')$, there is no guarantee about the relation between M and Decoder2(C').
In the worst case:

$$P(\text{error}_{\text{new}} \mid B) \leq 1.$$

From the optimal coupling result:

$$P(B) = P((M, C) \neq (M', C')) = \|P_{MC} - P_M P_C\|_{\text{TV}} \leq \delta,$$

$$P(A) = 1 - P(B) \leq 1.$$

Now combining the results and substituting back into the expression for the error probability, we get:

$$\begin{aligned} P(\text{error}_{\text{new}}) &\leq P(\text{error}_{\text{new}} \mid A) \cdot P(A) + P(\text{error}_{\text{new}} \mid B) \cdot P(B) \\ &\leq \varepsilon \cdot 1 + 1 \cdot \delta \\ &\leq \varepsilon + \delta. \end{aligned}$$

Thus, we have shown that:

$$P(\text{error}_{\text{new}}) \leq \varepsilon + \delta \qquad \text{Q.E.D.}$$

4 Additional topic: Coupling and convergence of Markov chains

4.1 **Objective:** observing an application of coupling in understanding the long-term behavior of an stochastic process.

4.2 Introduction

A Markov chain describes a system which moves stochastically between different states. Seldom, regardless of the initial state, the probability distribution of the chain converges to a unique stationary distribution π after a large number of steps. The coupling is a classic method of proving this convergence. Consider a simple two-state model of "weather": $S = \{\text{sunny}, \text{rainy}\}$. The transfer matrix is given by:

$$T = \begin{pmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{pmatrix}$$

The first row states that if it sunny today, then tomorrow will also be sunny with probability 0.9 and the seconds row states that if it is rainy today, then with a probability of 0.5, tomorrow will also be rainy.

4.3 To-Do

1. **Finding the stationary distribution:** Find the stationary distribution $\pi = (\pi_{\text{sunny}}, \pi_{\text{rainy}})$ by solving the linear equation $\pi T = \pi$ with the constraint $\pi_{\text{sunny}} + \pi_{\text{rainy}} = 1$.

If we define the event A as the event of sunny weather and event B as the event of rainy weather, we can also view the transfer matrix as

$$T = \begin{pmatrix} P(A|A) & P(B|A) \\ P(A|B) & P(B|B) \end{pmatrix}$$

It is easy to confirm that the sum of each row equals one (T is also called the Stochastic matrix).

$$(x \quad y)\begin{pmatrix} 0.9 & 0.1 \\ 0.5 & 0.5 \end{pmatrix} = (x \quad y) \rightarrow (9x + 5y \quad x + 5y) = 10(x \quad y)$$

Therefore we have 2 equations which are

$$x = 5y \quad ; \quad x + y = 1$$

Thus the answer is

$$\boxed{x = \frac{5}{6} \quad ; \quad y = \frac{1}{6}} \rightarrow \pi = (\frac{5}{6} \quad \frac{1}{6})$$

So we have found the steady state; this means after enough time the probability of this system to be in the sunny state is

$$\frac{5}{6}$$

And the probability of this system to be in the rainy state is

$$\frac{1}{6}$$

This result does not rely on the initial state. It depends only on stochastic matrix.

2. **Simulating a coupled process:** We simulate two versions of this weather process X_t and Y_t, which initially start in different states: $X_0 =$ sunny and $Y_0 =$ rainy. We couple their translations so that they "meet" as soon as possible.

- Implement a simulation for the pair (X_t, Y_t). In every step t, generate the next state (X_{t+1}, Y_{t+1}) with the following optimal coupling law:
  - If $X_t = Y_t$, then they have met each other and will move together from there onward (sample from the corresponding row of T).
  - If $X_t =$ sunny and $Y_t =$ rainy, generate a uniform random number U on $[0, 1]$.
    - If $U < 0.5$, set $X_{t+1} =$ rainy and $Y_{t+1} =$ rainy (they meet).
    - If $0.5 \leq U < 0.9$, set $X_{t+1} =$ sunny and $Y_{t+1} =$ rainy.
    - If $0.9 \leq U$, set $X_{t+1} =$ sunny and $Y_{t+1} =$ sunny (they meet).
- Repeat the simulation 10000 times and for each simulation, store the "meeting time" $T_{\text{meet}} = \min\{t | X_t = Y_t\}$. Plot the histogram of the meeting times.

We start with the given initial state:

$$\vec{X_0} = (1 \quad 0)^T \quad \text{: sunny} \qquad \vec{Y_0} = (0 \quad 1) \quad \text{: rainy}$$

So at moment t the vectors evolve as :

$$\vec{X_t} = T^t \vec{X_0} \qquad \vec{Y_t} = T^t \vec{Y_0}$$

In the simulation we showed that these vectors, after spending a moderate amount of time, converge to the eigenvector of T. In our simulation we assumed:

$X_t = (\text{component} \quad 0, \text{component} \quad 1)^T \qquad Y_t = (\text{component} \quad 2, \text{component} \quad 3)^T$

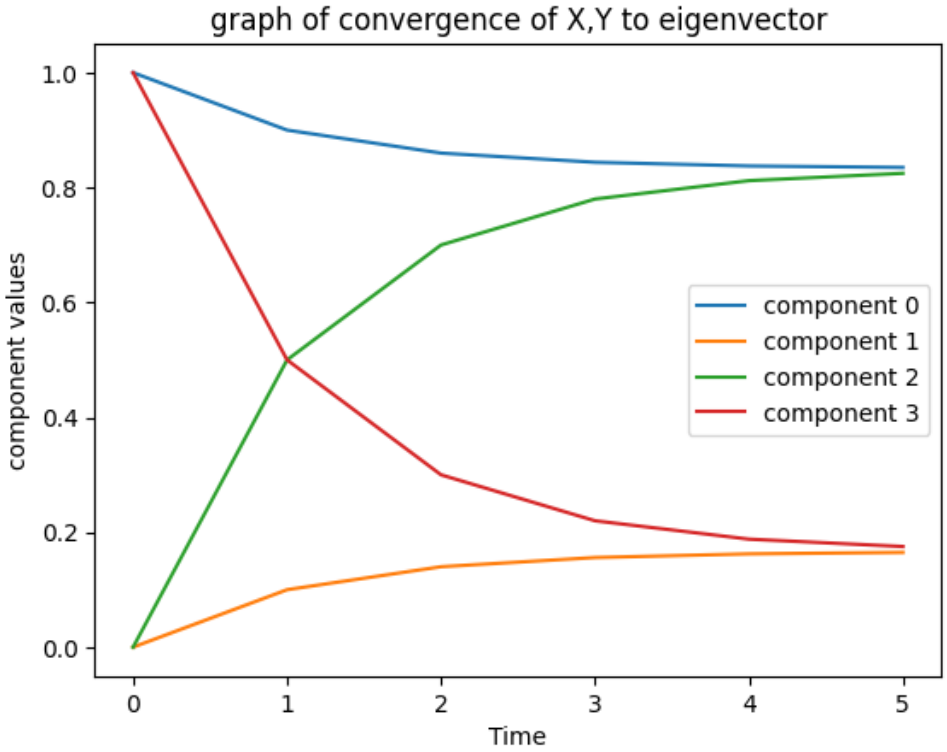


Figure 2: The convergence to the stationary state for different initial states

As we can see, components 0 and 2 tend to the value of 0.83 and components 1 and 3 tend to the value of 0.17, which was expected from our previous calculations.

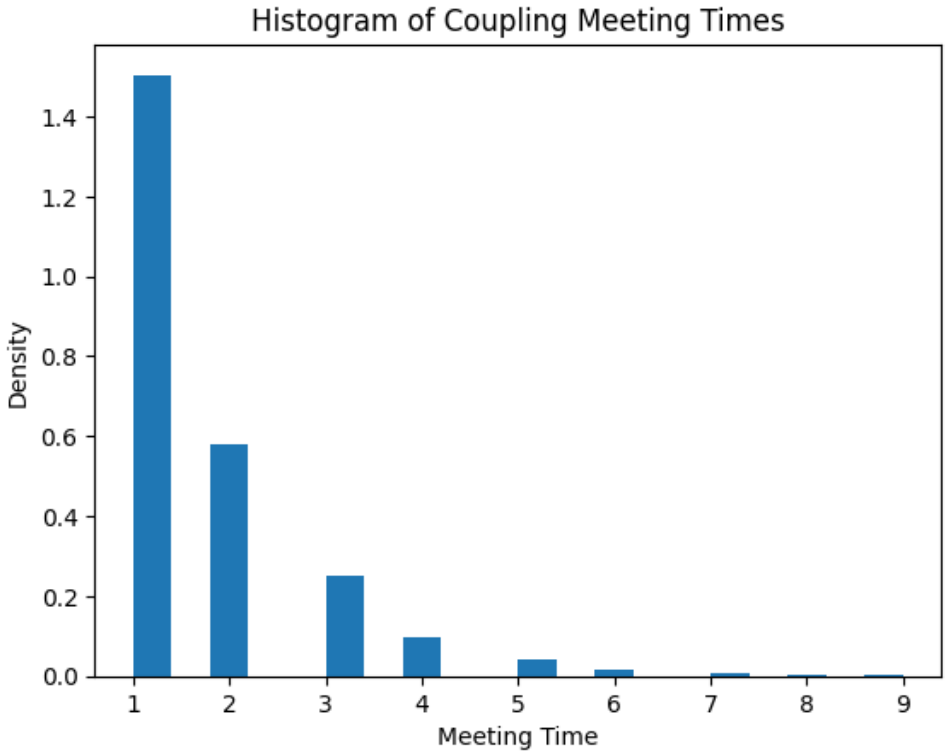


Figure 3: The histogram of the meeting times

3. **Connection to coupling:** The core idea is that the rate of intersection of the two chains, represents the rate of "forgetting" the initial state by the Markov chain. Explain, in your own terms, why a small expected meeting time, shows that the Markov chain converges rapidly to its stationary state.

   If the expected meeting time $\mathrm{E}[T_{\text{meet}}]$ is small, then with a high probability, the two chains become identical rapidly. This means that, After a short time, the distribution of X_t becomes almost independent of its initial state Because, regardless of your starting point, the chain quickly couples to the same trajectory. If the two chains, starting from different states, quickly become indistinguishable, then the chains must have "forgotten" their starting point; therefore the resulting distribution must be close to the stationary distribution.

   $$\text{Small expected meeting time} \Rightarrow \text{fast mixing} \Rightarrow \text{fast convergence to } \pi.$$

Appendix: Jupyter Notebook for Simulations

Required Libraries

In [22]:
```
import numpy as np
from matplotlib import pyplot as plt
```

Part 1: How different are two distributions?

In [ ]:
```
# defining the required function

def calculate_tv_distance(p, q):
    diff = np.array(p) - np.array(q)
    return np.sum(diff[diff > 0])

p = [0.1 , 0.2 , 0.3 , 0.4]
q = [0.4 , 0.3 , 0.2 , 0.1]
print(calculate_tv_distance(p, q))
```

```
0.4
```

Part 2: Building a bridge using Coupling

In [ ]:
```
# defining the functions and the required variables

def sample_from_optimal_coupling(p, q, n_samples):
    p = np.array(p)
    q = np.array(q)
    values = np.arange(len(p))

    C = np.sum(np.minimum(p, q))
    pq_min = np.minimum(p, q)
    matched_dist = pq_min / C

    p_rem = (p - pq_min) / (1 - C)
    q_rem = (q - pq_min) / (1 - C)

# initializing the arrays
    X = []
    Y = []

    for _ in range(n_samples):
        U = np.random.rand()
        if U < C:
            z = np.random.choice(values, p=matched_dist)
            X.append(z)
            Y.append(z)
        else:
            x = np.random.choice(values, p=p_rem)
            y = np.random.choice(values, p=q_rem)
            X.append(x)
            Y.append(y)

    return np.array(X), np.array(Y)

# the given distributions
```

```
p = [0.1, 0.2, 0.3, 0.4]
q = [0.4, 0.3, 0.2, 0.1]

# generating samples from the distributions
X, Y = sample_from_optimal_coupling(p, q, 10000)

# count (x != y) and divide by the size of n_samples
prob_neq = np.mean(X != Y)
print(f"Empirical P(X≠Y) = {prob_neq:.4f}")

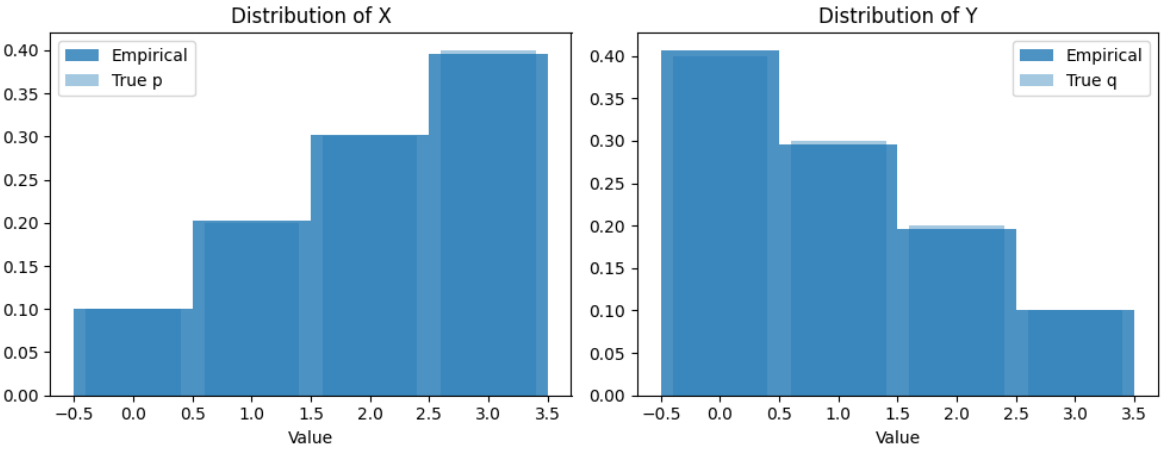
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

# histogram of X
axes[0].hist(X, bins=np.arange(5)-0.5, density=True, alpha=0.8, label='Empirical
axes[0].bar(range(len(p)), p, alpha=0.4, label='True p')
axes[0].set_title('Distribution of X')
axes[0].set_xlabel('Value')
axes[0].legend()

# histogram of Y
axes[1].hist(Y, bins=np.arange(5)-0.5, density=True, alpha=0.8, label='Empirical
axes[1].bar(range(len(q)), q, alpha=0.4, label='True q')
axes[1].set_title('Distribution of Y')
axes[1].set_xlabel('Value')
axes[1].legend()

plt.tight_layout()
plt.show()
```

```
Empirical P(X≠Y) = 0.4009
```



Part 3: From strong secrecy to perfect secrecy

In [ ]:
```
# No code for this part.
```

Part 4: Coupling and convergence of Markov chains

In [26]:
```
T = np.array([[0.9, 0.5],
              [0.1, 0.5]])

X0 = np.array([[1], [0]])
Y0 = np.array([[0], [1]])
```

```
XY_t = [[1], [0], [0], [1]]

N = 5
for _ in range(N):
    X0, Y0= T @ X0, T @ Y0
    XY_t[0].append(X0[0,0])
    XY_t[1].append(X0[1,0])
    XY_t[2].append(Y0[0,0])
    XY_t[3].append(Y0[1,0])

t = range(N+1)
for i in range(4):
    plt.plot(t, XY_t[i], label=f'component {i}')

plt.legend()
plt.xlabel("Time")
plt.ylabel("component values")
plt.title("graph of convergence of X,Y to eigenvector")
plt.show()

#-----------------------------------------------------------------------

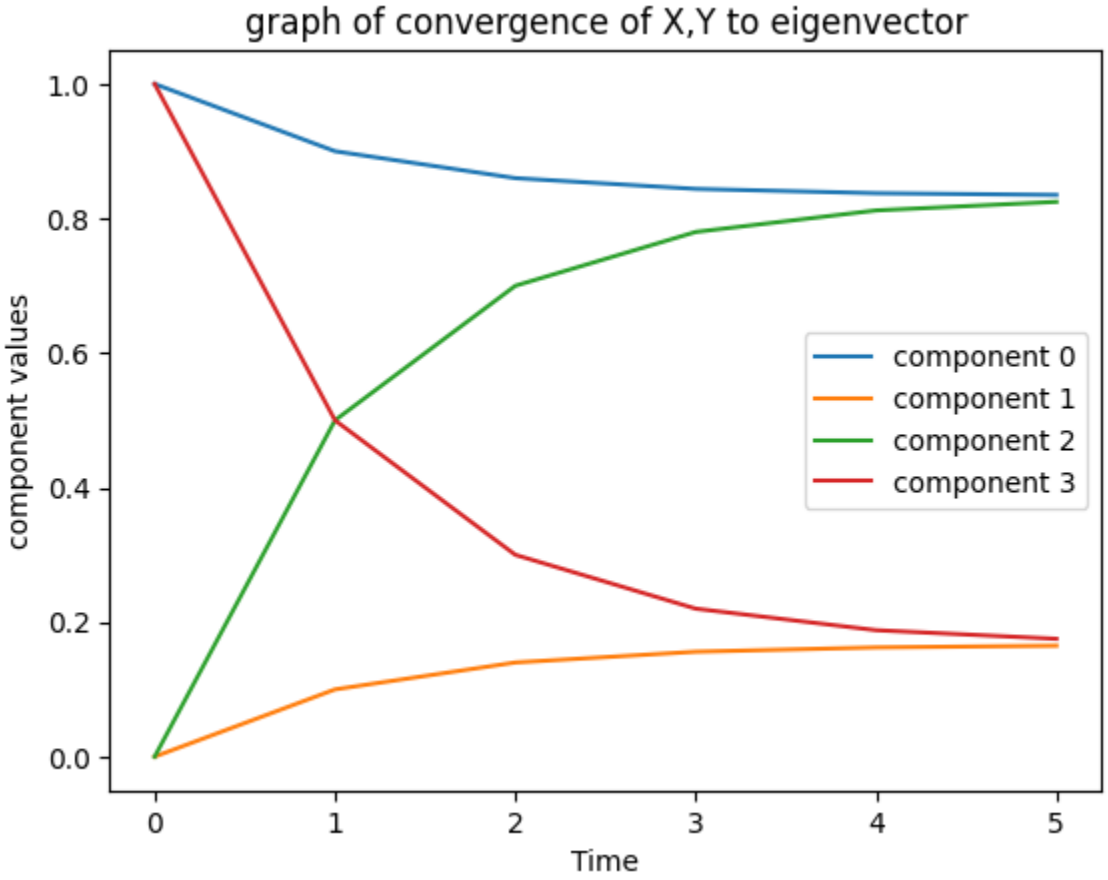
# 0 = Sunny, 1 = Rainy

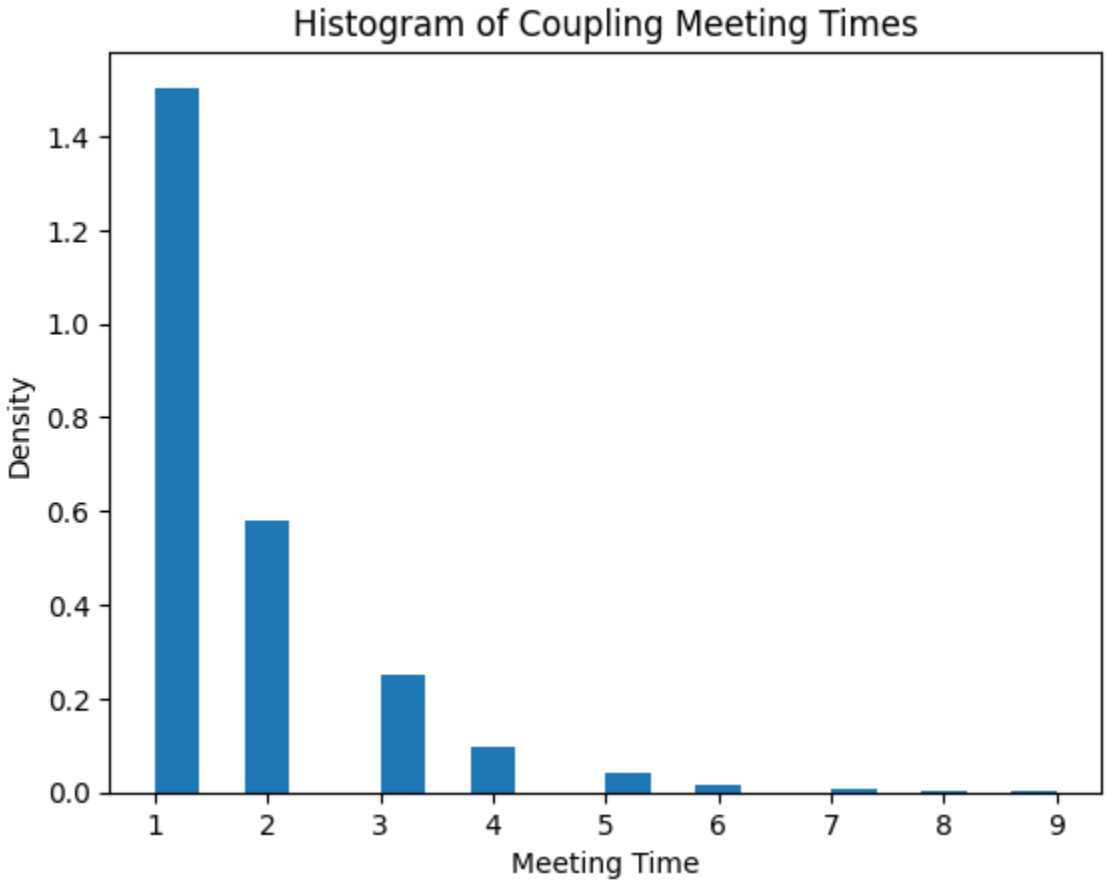
def simulate_meeting_time():
    X = 0
    Y = 1
    t = 0
    while X != Y:
        U = np.random.rand()
        if U < 0.5:
            X, Y = 1, 1
        elif 0.5 <= U < 0.9:
            X, Y = 0, 1
        else:
            X, Y = 0, 0
        t += 1
    return t

N = 10000
meeting_times = [simulate_meeting_time() for _ in range(N)]

plt.hist(meeting_times, bins=20, density=True)
plt.xlabel("Meeting Time")
plt.ylabel("Density")
plt.title("Histogram of Coupling Meeting Times")
plt.show()

print("Average meeting time:", np.mean(meeting_times))
```






```
Average meeting time: 1.6772
```